DIPAK SUMESARA

Senior Backend Engineer

Node.js | TypeScript | AWS | Microservices | PostgreSQL, MongoDB, Redis

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PROFESSIONAL SUMMARY

Senior backend engineer with 8 years building scalable, event-driven Node.js and TypeScript services on AWS. Sole backend owner of Aloha at SwagUp, a multi-tenant US B2B commerce platform (clients include Walmart, Nasdaq and Logitech): cut core API latency from 2-3 s to ~200 ms, held 99.9% uptime, and ran 4 AWS Lambda microservices, a Salesforce integration at 5K+ messages a day and 50K-record batch migrations. Technical lead who has mentored engineers and set code-review and testing standards, with full-stack range in React and Angular. Earlier: a real-time backend for 100K+ concurrent users, services for Singapore's national health app (1M+ downloads), and a modular rewrite whose package model tripled revenue.

TECHNICAL SKILLS

Backend: Node.js, TypeScript, JavaScript, Express.js, NestJS, REST and GraphQL API design, API versioning, webhooks, WebSockets, OAuth 2.0, JWT

Architecture: Microservices, cloud-native, serverless, distributed systems, event-driven architecture, pub/sub, asynchronous processing, message queues (AWS SQS, BullMQ), multi-tenant design, system design, OOP, SOLID, design patterns, data structures and algorithms

Data: PostgreSQL, MongoDB (Mongoose), MySQL, Redis, Elasticsearch; SQL and NoSQL schema design, indexing, query optimization, caching

Cloud & DevOps: AWS (Lambda, API Gateway, SQS, S3, CloudFront), Docker, Kubernetes, CI/CD (GitHub Actions), PM2, Git

Reliability & observability: Monitoring and alerting, Sentry, structured logging, incident response, root-cause analysis, 99.9% uptime, rate limiting

Security & quality: RBAC, OWASP Top 10, CORS configuration, secure coding, input validation, automated testing (Jest, Vitest), unit and integration testing, ESLint, SonarQube, code review, mentoring, Agile/Scrum

Frontend: React, Redux, React Query, Material UI, Angular, Ionic

AI: LLM integration (Claude API, RAG, tool calling, Zod); AI-assisted development with Claude Code, Cursor, GitHub Copilot, OpenAI Codex

PROFESSIONAL EXPERIENCE

Independent: open-source backend and LLM projects (see Projects) **Apr 2026 - Present**

SwagUp | Remote, US team **Mar 2023 - Apr 2026**

Senior Backend Engineer (promoted from Full Stack Developer) | *Aloha B2B platform: hundreds of vendors, thousands of orders a month; via Extendime*

- **Performance:** Cut core API latency from 2-3 s to ~200 ms (~90%) by profiling slow endpoints and applying query rewrites, targeted indexes and Redis caching of hot, read-heavy responses.
- **Architecture:** Split a backend where vendor, distributor and admin workflows competed into 4 AWS Lambda microservices behind API Gateway with versioned REST and GraphQL APIs, schema validation and response transformers; moved cross-service side effects to pub/sub events and SQS workers with retries and idempotent handlers, reducing the impact of slow third-party integrations on user-facing requests.
- **Integrations:** Built the Salesforce integration on webhooks and asynchronous SQS processing (5K+ messages a day) to keep Aloha, swagup.com and Salesforce eventually consistent, with a dead-letter queue for failed-message investigation and recovery.
- **Data migration:** Built scheduler-triggered batch migrations on Node.js worker threads (up to 50K records per run) with controlled chunked writes and idempotent upserts, so reruns were safe; runs completed without server crashes, database load spikes or data loss.
- **Identity and data:** Built the PostgreSQL identity service every microservice trusts (JWT, API keys, Google OAuth, RBAC, sessions); kept vendor, distributor, product and order data in MongoDB.
- **Reliability and observability:** Held 99.9% uptime under HA SLAs (connection pooling, read/write splitting, failover) with monitoring, structured logging and alerting; applied OWASP-aligned API protections; led incident response from Sentry, logs and alerts to root cause, rollback or hotfix.
- **Commerce and search:** Built swagup.com's pricing engine, payment-gateway integration in the order flow and order-tracking APIs, plus Elasticsearch search over ~20K products kept consistent with MongoDB.
- **Full-stack and standards:** Started as a Full Stack Developer building Aloha's admin, quote and pricing screens in React (Redux, React Query, Material UI). Set the team's Jest testing, ESLint, SonarQube and code-review standards; shipped via Docker, Kubernetes and GitHub Actions; used Claude Code, Cursor, GitHub Copilot and OpenAI Codex under the same review bar as hand-written code.

CoffeeBeans Consulting | Remote **Jun 2021 - Mar 2023**

Software Engineer / Backend Lead

- **Propsoch:** Technical lead and the client's engineering contact for Propsoch, a Bengaluru homebuyer-advisory platform; built Google OAuth sign-in, property ranking from Google Maps distance calculations, and a procurement workflow where users complete large forms incrementally without losing entered values.
- **Salam Kisan:** Owned backend architecture on Strapi CMS (REST APIs, Open edX integration, data-sync workflows on Redis-backed BullMQ job queues); led a 3-engineer backend team through sprints, design and code reviews, and production support.
- **Healthy 365 (Singapore Health Promotion Board):** Built Node.js and TypeScript services on PostgreSQL and MongoDB for Singapore's national health app (1M+ downloads) in a 10-person team, under a 90% unit-test coverage gate with strict ESLint rules and SonarQube analysis, with CORS configuration, input validation and SQL injection, XSS and CSRF protections.
- Integrated payment gateways into client checkout flows.

Yudiz Solutions | Ahmedabad, India **May 2020 - May 2021**

Software Engineer

- **11Wickets:** Built a real-time backend for a fantasy sports platform supporting 100K+ concurrent users at peak match load, using WebSockets for live scores and chat and Redis caching of hot match data to reduce PostgreSQL read pressure.
- Integrated the payment gateway for contest entry; worked on authentication, secure APIs and performance; ran services under PM2 with CI/CD; mentored junior developers through code reviews.

Biztech Consulting & Solutions | Ahmedabad, India **Jan 2019 - Mar 2020**

Software Engineer

- **BrushYourIdeas 2.0:** Proposed and led the rewrite from legacy Angular to independent feature modules with clean design patterns and secure API integrations; Professional, Ultimate and Custom packages were assembled from those modules, and the custom-package model tripled revenue.

Angular Minds | Pune, India **Dec 2017 - Sep 2018**

Software Engineer (intern, converted to full-time)

- **Numnu:** Built the NestJS and MongoDB backend (proximity and date-range event search) and the Ionic app for a food-event discovery product, launched on Google Play and the Apple App Store; its event-guide mode was used at the Royal Manitoba Winter Fair.

SELECTED OPEN-SOURCE PROJECTS

Distributed Rate Limiter (Fastify, Redis Lua, Prometheus, k6): token bucket shared by stateless API instances through one atomic Lua script. github.com/dipak-sumesara/high-performance-api-engine

AI Integration Discovery Agent (Claude API, Ollama, Zod, Vitest): tool-calling agent behind one provider interface, validated output, SSRF and path-traversal guards, 18 tests. github.com/dipak-sumesara/ai-integration-discovery-agent

AI Commerce Assistant (React, Fastify, RAG, SSE, Zod, Vitest): streaming retrieval-augmented shopping assistant in an npm-workspaces monorepo with shared Zod contracts, a scope classifier and prompt-injection tests (15 tests). github.com/dipak-sumesara/ai-commerce-assistant

EDUCATION

Master of Computer Applications (MCA), Savitribai Phule Pune University **2018**

Bachelor of Computer Applications (BCA), Krantiguru Shyamji Krishna Verma Kachchh University **2015**